

Microservices

Part 1: Introduction and Decomposition Pattern

Introduction

- This is part 3 of the video series on high-level design
- Previous 2 videos already covered
- This part covers:
 - Monolithic vs microservices
 - Covering important microservices patterns
 - Says this topic can lead to around 15 questions in interviews

Disadvantages of Monolithic Architecture

- IDE Overloading
- Tight Coupling
 - Changing one line can impact other components
 - Need to test/deploy entire application for one change
- Difficult to scale
 - If one component needs scaling, entire application needs to be scaled
- Expensive deployments and rollbacks
 - Entire application needs redeployment for a small change
- Large codebase
 - Everything in a single application
 - Codebase grows large over time
 - Difficult to make changes, understand impact

Why Microservices?

- To overcome monolithic disadvantages
- Split large application into small services

Advantages of Microservices

- Better separation of concerns
- Loose coupling between services
- Independent deployment of services
- Easy to scale out specific services
- Faster release cycles

Disadvantages of Microservices

- Proper service boundaries/decomposition is challenging
- Inter-service communication is complex
 - Monitoring calls across services
 - Handling failures
- Distributed transaction management is difficult
 - Across multiple databases

Microservices Design Phases

- Decomposition patterns
 - By business capability

- By subdomain - Domain Driven Design (DDD)
- Database patterns
 - DB per service
 - Shared DB
- Communication patterns
 - via API
 - via Events
- Integration patterns
 - API gateways etc.
- Deployment patterns
- Cross-cutting concerns like monitoring, logging, observability

Decomposition Patterns

- By business capability
 - Split based on business functions like order mgmt, inventory, etc
 - requires good business knowledge
- By subdomain - Domain Driven Design (DDD)
 - Split large domains into multiple services
 - E.g. splitting payment domain into forward and refund payments